Adult, Spouse & Child Age Validation

Final Functional / PRD-Level Specification

1. Field Requirement

Field	Required / Optional	Rule
Adult Age	Mandatory	Must always be provided and validated.
Spouse Age	Optional	If provided, all spouse-related age validations apply.
Child 1 Age	Optional	If provided, all applicable child validations apply.
Child 2 Age	Optional	If provided, all applicable child validations apply.
Child 3 Age	Optional	If provided, all applicable child validations apply.
Child 4 Age	Optional	If provided, all applicable child validations apply.

Important: Optional age fields may be null/empty. Once an optional age is provided, every applicable relationship validation must execute.

2. Adult Age Validation

- Adult Age is mandatory.
- Adult Age must be a valid whole number.
- Adult Age must be at least 18 years.

Scenario	Result	Backend Message
Adult age missing	Invalid	age is required
Adult age is not a whole number	Invalid	age must be a valid whole number
Adult age < 18	Invalid	age must be at least 18
Adult age >= 18	Valid	-

3. Spouse Age Validation

Spouse Age is optional. If it is empty/null, spouse-age validation is skipped. If it is provided, it must be a whole number and at least 18 years old.

Adult	Spouse	Result	Reason
40	Empty	Valid	Spouse is optional
40	17	Invalid	Spouse age is below 18
40	18	Valid	Minimum spouse age satisfied
40	35	Valid	Spouse age is valid

Backend messages:

- spouse_age must be a valid whole number
- spouse_age must be at least 18

4. Child Age Basic Validation

Child 1, Child 2, Child 3 and Child 4 are optional. Any provided Child Age must be a valid whole number and cannot be negative.

There is no fixed maximum Child Age. A child may be older than 25 if the required 21-year parent-child age gap is maintained.

Child Age	Result	Reason
Empty	Valid	Child age is optional
-1	Invalid	Child age cannot be negative
0	Valid	Non-negative age
26	Conditionally Valid	Depends on Adult/Spouse 21-year gap
34	Conditionally Valid	Depends on Adult/Spouse 21-year gap

Backend messages:

- child_N_age must be a valid whole number
- child_N_age cannot be negative

5. Adult-to-Child Age Gap

Every provided child must be at least 21 years younger than the Adult.

Formula: Adult Age - Child Age >= 21

Adult	Child	Difference	Result
55	26	29	Valid
55	34	21	Valid
55	35	20	Invalid
40	19	21	Valid
40	20	20	Invalid
30	9	21	Valid
30	10	20	Invalid

Backend message: parent age must be at least 21 years greater than child_N_age

6. Spouse-to-Child Age Gap

This validation runs only when BOTH Spouse Age and the respective Child Age are provided.

Formula: Spouse Age - Child Age >= 21

Adult	Spouse	Child	Adult Gap	Spouse Gap	Result
55	Empty	30	25	Skipped	Valid
55	50	29	26	21	Valid

55	50	30	25	20	Invalid

Backend message: spouse age must be at least 21 years greater than child_N_age

7. Dynamic Maximum Child Age

There is no fixed maximum child age.

Without Spouse: Maximum Child Age = Adult Age - 21

With Spouse: Maximum Child Age = MIN(Adult Age, Spouse Age) - 21

Adult	Spouse	Maximum Allowed Child Age
55	Empty	34
55	50	29
45	40	19
30	28	7

8. Child-to-Child Age Gap

When two or more Child Ages are provided, every pair of provided children must have at least a 2-year age difference.

Formula: ABS(Child A Age - Child B Age) >= 2

When all four children are provided, validate these six pairs:

- Child 1 ↔ Child 2
- Child 1 ↔ Child 3
- Child 1 ↔ Child 4
- Child 2 ↔ Child 3
- Child 2 ↔ Child 4
- Child 3 ↔ Child 4

Child 1	Child 2	Child 3	Child 4	Result	Reason
20	18	16	14	Valid	All pair differences are >= 2
20	19	16	14	Invalid	Child 1 and Child 2 gap is 1
20	18	17	14	Invalid	Child 2 and Child 3 gap is 1
20	20	16	14	Invalid	Duplicate/same age gives gap 0

Backend message: minimum age difference between children must be 2 years

9. Optional Field Scenarios

Adult	Spouse	Children	Result	Validation Applied
40	Empty	Empty	Valid	Adult only

40	38	Empty	Valid	Adult + Spouse
40	Empty	19	Valid	Adult-child gap = 21
40	Empty	20	Invalid	Adult-child gap = 20
55	Empty	26	Valid	Adult-child gap = 29
55	Empty	34	Valid	Adult-child gap = 21
55	Empty	35	Invalid	Adult-child gap = 20
55	50	Empty	Valid	Children optional
55	50	29	Valid	Both parent gaps >= 21
55	50	30	Invalid	Spouse-child gap = 20
50	Empty	20, 18	Valid	Adult gaps valid; child gap = 2
50	Empty	20, 19	Invalid	Child gap = 1
50	47	20, 18, 16, 14	Valid	All validations pass
50	47	20, 19, 16, 14	Invalid	Child 1/2 gap = 1

10. Non-Sequential Optional Child Fields

Because each Child Age is optional, validation should be based on populated fields, not on the child index.

Child 1	Child 2	Child 3	Child 4	Expected Handling
20	Empty	16	Empty	Validate Child 1 and Child 3 only, including their 4-year gap
Empty	18	Empty	14	Validate Child 2 and Child 4 only, including their 4-year gap
Empty	Empty	16	Empty	Validate Child 3 against Adult and Spouse (if provided)
Empty	Empty	Empty	Empty	No child validation

11. Validation Execution Order

1. Validate Adult Age: required, whole number, and >= 18.
2. If Spouse Age is provided: validate whole number and >= 18.
3. For each provided Child Age: validate whole number and non-negative value.
4. For each provided Child Age: validate Adult Age - Child Age >= 21.
5. If Spouse Age is provided: validate Spouse Age - each provided Child Age >= 21.
6. If two or more Child Ages are provided: validate every populated child pair has an absolute age difference >= 2.

12. Final Business Rules

Rule	Final Validation
Adult	Mandatory; whole number; Adult Age >= 18
Spouse	Optional; if provided, whole number and Spouse Age >= 18

Child 1-4	Optional; if provided, whole number and Child Age >= 0
Adult vs Child	Adult Age - Child Age >= 21
Spouse vs Child	If spouse is provided: Spouse Age - Child Age >= 21
Child vs Child	For every provided pair: ABS(Child A Age - Child B Age) >= 2
Fixed maximum child age	Not applicable

13. Backend Error Message Matrix

Validation Failure	Backend Message
Adult missing	age is required
Adult invalid	age must be a valid whole number
Adult below 18	age must be at least 18
Spouse invalid	spouse_age must be a valid whole number
Spouse below 18	spouse_age must be at least 18
Child invalid	child_N_age must be a valid whole number
Child negative	child_N_age cannot be negative
Adult-child gap < 21	parent age must be at least 21 years greater than child_N_age
Spouse-child gap < 21	spouse age must be at least 21 years greater than child_N_age
Child-child gap < 2	minimum age difference between children must be 2 years

14. Final Decision Logic

Adult Age is mandatory. Spouse Age and all four Child Ages are optional.

If an optional field is absent, validations depending on that field are skipped. If it is present, all applicable validations become mandatory.

Without spouse: Child Age <= Adult Age - 21

With spouse: Child Age <= MIN(Adult Age, Spouse Age) - 21

For multiple provided children: ABS(Child A Age - Child B Age) >= 2 for every pair.